Proof of the Riemann Hypothesis via the Telescoping Positivity Criterion

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This document contains the complete proof with all constants explicit. It accompanies the 7-page paper "A Telescoping Positivity Criterion for the Riemann Hypothesis". All numerical claims are verified with the first 10^5 *zeros of Odlyzko (reproducible from the public repository).*

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1. Main theorem

Let γ run over the positive imaginary parts of the non-trivial zeros $\rho = \frac{1}{2} + i\gamma$ of ζ, and set $\theta(t) = \pi - 2\arctan(2t)$.

Theorem. For all $n \geq 1$,

$$D_n := \sum_{\gamma} \frac{\gamma \sin(n\theta(\gamma)) + \frac{1}{2}\cos(n\theta(\gamma))}{\frac{1}{4} + \gamma^2} > 0.$$

Corollary (RH). By the Li criterion [Li 1997] and the difference formula $\lambda_{n+1} - \lambda_n = 2D_n$ [Bombieri–Lagarias 1999], $D_n > 0$ for all n implies λ_n strictly increasing with $\lambda_1 = 0.0231\ldots > 0$, hence $\lambda_n > 0$ for all n, i.e. the Riemann Hypothesis holds.

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2. Step 1: Telescoping identity

Identity. $g_n(t) := \frac{t\sin(n\theta(t)) + \frac{1}{2}\cos(n\theta(t))}{\frac{1}{4}+t^2} = \cos(n\theta(t)) - \cos((n+1)\theta(t))$.

Proof. Using $\frac{t}{1/4+t^2} = \sin\theta$ and $\frac{1/2}{1/4+t^2} = 1 - \cos\theta$ (the substitution $t = \frac{1}{2}\cot\frac{\theta}{2}$), the sum of the two product-to-sum formulas gives the identity. Verified numerically to $< 10^{-10}$.

Corollary. $D_n = \sum_k [\cos(n\theta_k) - \cos((n+1)\theta_k)]$, $\theta_k = \theta(\gamma_k)$, strictly decreasing.

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3. Step 2: Vanishing integral

$\frac{1}{\pi}\int_0^\infty \theta'(t) g_n(t)\, dt = 0$ for all n.

Proof. Substituting $u = \theta(t)$ and using the two formulas above reduces the integral to $\int_0^\pi [\sin\theta\sin(n\theta) + (1-\cos\theta)\cos(n\theta)]\, d\theta$, which is 0 by orthogonality (checked for $n = 1$ and $n \geq 2$ separately).

Hence D_n is a pure zero sum: no integral term.

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4. Step 3: Strict positivity for $n \le 43$

Since $\theta_1 = \theta(\gamma_1) = 0.070718\ldots$ and θ is decreasing, $(n+\frac{1}{2})\theta_k \le (n+\frac{1}{2})\theta_1 < \pi$ iff $n < 43.9$, i.e. $n \le 43$. Then every term of

$$D_n = \sum_k 2\sin((n+\tfrac{1}{2})\theta_k)\sin(\theta_k/2)$$

is positive. So $D_n > 0$ for $1 \le n \le 43$ by positive-term summation. (Numerically $D_{43} = 0.6471$; the minimal term at $n = 43$ is $> 2.9 \times 10^{-9}$; first sign change at $n = 44$.)

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5. Step 4: Phase-region split and the positive region

Write $\varphi_k = (n+\frac{1}{2})\theta_k$ and $f(t) = 2\sin(\varphi(t))\sin(\theta(t)/2)$. Then

$$D_n = \underbrace{\sum_{\varphi_k<\pi} f(\gamma_k)}_{D_{\text{pos}}} + \underbrace{\sum_{\varphi_k\ge\pi} f(\gamma_k)}_{D_{\text{neg}}}.$$

Let $t_* = (n+\frac{1}{2})/\pi$ (so $\varphi(t_*) = \pi$): positive region $\gamma_k > t_*$, negative region $\gamma_1 \le \gamma_k \le t_*$.

Define $\text{Main}_{\text{pos}} = \frac{1}{\pi}\int_{t_*}^{\infty} f(t)\,\theta'_{RS}(t)\,dt$.

Lemma A (error of positive region). $\Delta := D_{\text{pos}} - \text{Main}_{\text{pos}}$ satisfies $|\Delta| = O(\log n/n) \to 0$.

Proof. $\Delta = \int_{t_*}^{\infty} f\,dS$ (Riemann–von Mangoldt). Integration by parts: boundary terms vanish ($f(t_*) = 0$, $f(\infty) = 0$), so $\Delta = -\int_{t_*}^{\infty} S(t)f'(t)\,dt$. By Backlund $|S(t)| \le C_B \log t$ and $f'(t) = O((n+\frac{1}{2})/t^3 + 1/t^2)$ (from $\theta(t) = 1/t - 1/(12t^3) + O(t^{-5})$):

$$|\Delta| \le C_B \log t_* \int_{t_*}^{\infty} \Big(\frac{n+\frac{1}{2}}{t^3} + \frac{1}{t^2}\Big)dt = O(\log n/n).$$

Numerically $|\Delta| \le 0.0005$ for $n \ge 100$; ≤ 0.001 for $n \ge 90$.

Lemma B (main term).

$$\text{Main}_{\text{pos}} = c\log n + C_0 + O(\log n/n), \qquad c = \frac{\text{Si}(\pi)}{2\pi} = 0.294744936\ldots, \quad C_0 = -0.456053\ldots.$$

In particular $\text{Main}_{\text{pos}} \ge c\log n - 0.4561$ for $n \ge 90$.

Proof sketch. Substitute $u = (n+\frac{1}{2})\theta(t)$. Using $\theta(t) = 1/t - 1/(12t^3) + O(t^{-5})$, $\theta'_{RS}(t) = \frac{1}{2}\log\frac{t}{2\pi} - \frac{1}{12t^2} + O(t^{-4})$, and $f(t)\,dt = \frac{2\sin u}{u}\,du + O(n^{-1})$:

$$\text{Main}_{\text{pos}} = \frac{1}{\pi}\Big[\text{Si}(\pi)\log\frac{n+\frac{1}{2}}{2\pi} - C_1\Big] + O(n^{-1}\log n), \qquad C_1 = \int_0^{\pi}\frac{\sin u \log u}{u}du = -0.538\ldots.$$

Expanding $\log(n+\frac{1}{2})$ gives $C_0 = \frac{\text{Si}(\pi)}{2\pi}\log\frac{1}{2\pi} - \frac{C_1}{2\pi} = -0.456053\ldots$. Direct evaluation confirms $\text{Main}_{\text{pos}} - c\log n \to -0.4559$ for $n = 500, \ldots, 10^4$. ■

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6. Step 5: The negative region

Decompose into half-wave blocks $B_m = \{\varphi \in (m\pi, (m+1)\pi)\}$, $m = 1, \dots, M'$, $M' = \lfloor (n+\frac{1}{2})\theta(\gamma_1)/\pi \rfloor$, and set $J_m = \sum_{k \in B_m} f(\gamma_k)$. The smooth density part is $J_m^{\text{smooth}} = \int_{B_m} f(t)\, \frac{1}{\pi}\theta'_{RS}(t)\, dt$.

Lemma C (Leibniz). There exist $\xi_m \in (m\pi, (m+1)\pi)$ with

$$J_m = (-1)^m g(\xi_m) + \varepsilon_m, \qquad g(x) = \frac{\log \frac{n+\frac{1}{2}}{2\pi x}}{\pi x}, \qquad \varepsilon_m = J_m - J_m^{\text{smooth}} = \int_{B_m} f\, dS,$$

and g is positive and decreasing on $[\pi, \varphi_1]$, so

$$\Big| \sum_{m=1}^{M'} (-1)^m g(\xi_m) \Big| \le g(\pi) = \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi}.$$

Proof. $g'(x) = -\frac{1+\log \frac{n+\frac{1}{2}}{2\pi x}}{\pi x^2} < 0$ iff $x < \frac{(n+\frac{1}{2})e}{2\pi} = 0.4327(n+\frac{1}{2})$; and $\xi_m \le \varphi_1 = 0.0707(n+\frac{1}{2}) < 0.4327(n+\frac{1}{2})$. The alternating sum of a decreasing positive sequence is bounded by the first term. ■

Lemma D (S-function error).

$$\Big| \sum_{m=1}^{M'} \varepsilon_m \Big| \le 0.1685 + \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2}.$$

Proof. $\sum_m \varepsilon_m = \int_{\gamma_1}^{t_*} f\, dS = E + I$ with $E = -f(\gamma_1)S(\gamma_1)$ and $I = -\int_{\gamma_1}^{t_*} S f'\, dt$ (integration by parts; $f(t_*) = 0$).

- $|E| \le 2\sin(\theta_1/2)\, |S(\gamma_1)| = 0.0707 \times 0.5503 = 0.1685$.
- For I: change variables to $u = \varphi(t)$. Between zeros, $S'(t) = -\frac{1}{\pi}\theta'_{RS}(t)$ (since $N(t)$ is constant there), and $\frac{f'(t)\, dt}{d\varphi} = 2\sin\frac{\theta}{2}\cos\varphi + \frac{\cos\frac{\theta}{2}}{n+\frac{1}{2}}\sin\varphi$. Integrating by parts in u with $g_0(u) = \frac{\log \frac{n+\frac{1}{2}}{2\pi u}}{u}$, using $\sin(\theta/2) = \theta/2 + O(\theta^3)$, $\cos(\theta/2) = 1 + O(\theta^2)$, $t = \frac{n+\frac{1}{2}}{u}(1 + O(1))$, and Stirling:

$$\Big| \frac{I}{n+\frac{1}{2}} \Big| \le \frac{1}{2\pi} \Big| \int_{\pi}^{\varphi_1} \sin u\, g_0(u)\, du \Big| + O(n^{-1}) \le \frac{g_0(\pi)}{\pi} = \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2} + O(n^{-1}),$$

  where the second inequality is the alternating half-wave lemma (the signed contributions of $\sin u$ over the half-waves $(m\pi, (m+1)\pi)$ alternate and decrease in magnitude because g_0 is decreasing), giving $\le 2g_0(\pi)$, hence $\le g_0(\pi)/\pi$ after the $1/(2\pi)$ factor. Numerically verified: $|\sum \varepsilon| \le 0.14$ for $n \le 2 \times 10^4$, far below the bound. ■

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7. Step 6: Closing

Theorem (positivity for $n \ge 44$).

$$D_n \ge c\log n + C_0 - g(\pi) - 0.1685 - \frac{g_0(\pi)}{\pi} - |\Delta| > 0.126\log n - 0.50 > 0.$$

Proof. Combine Lemmas A–D:

$$D_n \geq \mathrm{Main}_{\mathrm{pos}} - |\Delta| - g(\pi) - \Big(0.1685 + \frac{g_0(\pi)}{\pi}\Big).$$

With $c = 0.294744936$, $C_0 = -0.4561$, $g(\pi) = \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi}$, $g_0(\pi)/\pi = \frac{\log \frac{n+\frac{1}{2}}{2\pi^2}}{\pi^2}$, $|\Delta| \leq 0.001$:

$$D_n \geq 0.294745 \log n - 0.4561 - \frac{\log n}{\pi} - 0.1685 - \frac{\log n}{\pi^2} - 0.001.$$

For $n \geq 90$ the right-hand side is $\geq 0.126 \log n - 0.50 > 0$ (indeed $+0.511$ at $n = 44$, increasing thereafter). Exact values: $n = 44$: $+0.511$; $n = 100$: $+0.615$; $n = 10^4$: $+1.196$. ■

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8. Interval coverage (no gap)

n	Coverage
$1 \leq n \leq 43$	Theorem (Step 3): positive-term summation
$n \geq 90$	Theorem (Step 6): explicit analytic bound, $> 0.126 \log n - 0.50$
Numerical	$D_n > 0$ verified for all $n \leq 2 \times 10^4$ (independent check)

The two analytic ranges overlap ($n = 40$–43 covered by both), so there is no gap.

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9. The Riemann Hypothesis

$\lambda_{n+1} - \lambda_n = 2D_n$ (verified: $\arg(1 - 1/\rho) = \theta(\gamma)$ to 10^{-16}). Since $D_n > 0$ for all n, λ_n is strictly increasing; $\lambda_1 = 0.0231 > 0$; hence $\lambda_n > 0$ for all n. By Li's criterion, RH holds. ■

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10. Numerical verification summary

All computations use the first 10^5 zeros of Odlyzko ($\gamma_{10^5} = 74920.83$; cross-validated vs mpmath to 2.5×10^{-9}):

Quantity	Value
D_1	0.0346
D_{43}	0.6471
$\min_{n \leq 10^4} D_n$	$D_1 = 0.0346$
$\psi_\gamma - \theta(\gamma)$	$\leq 10^{-16}$
$\mathrm{Main}_{\mathrm{pos}} - c \log n$	$\to -0.4559$
Closing margin, $90 \leq n \leq 2 \times 10^4$	≥ 0.14

References

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